FTC Shooting mechanism

# FTC Shooting and Intake Mechanism Design

In shooting, we require

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1) **Accuracy**

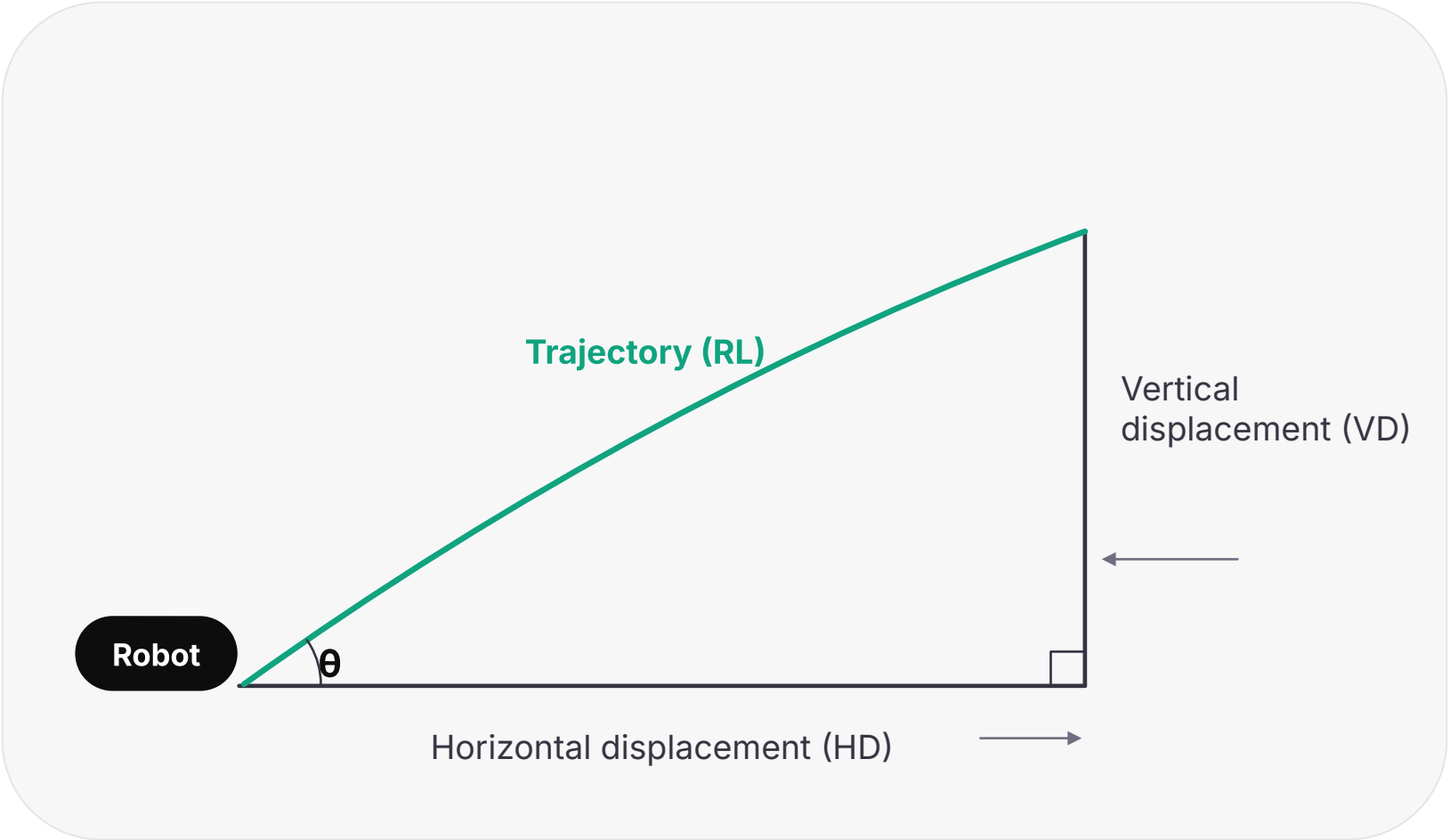
2) **Consistency**

3) **Speed**

$$\frac{\text{Number of balls}}{\text{time taken}}$$

"For consistency, we require all the **VARIABLES** to become **CONSTANTS**"

# Variables while shooting




## HD

1) Horizontal displacement

## VD

2) Vertical displacement

## RL

3) Trajectory (RL)

a) Speed at which the ball is shot (V)

b) Angle (θ)

01

## VD

As per FTC rules, it remains constant.

02

## Angle (θ)

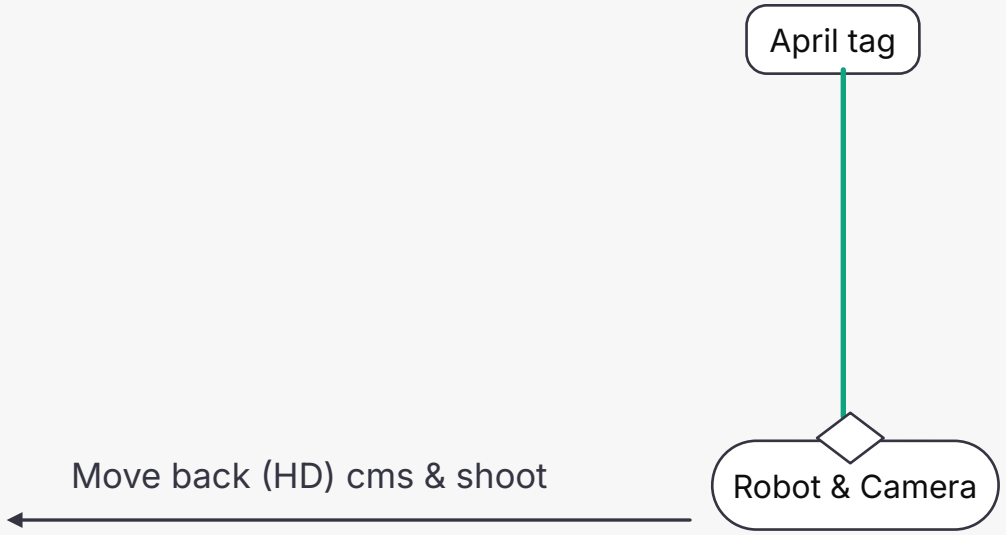
Because the robot has a fixed shooting mechanism, θ remains constant.

# What is already constant

Therefore, We have to make **(HD) & (V)** Constant!

# Making (HD) Constant

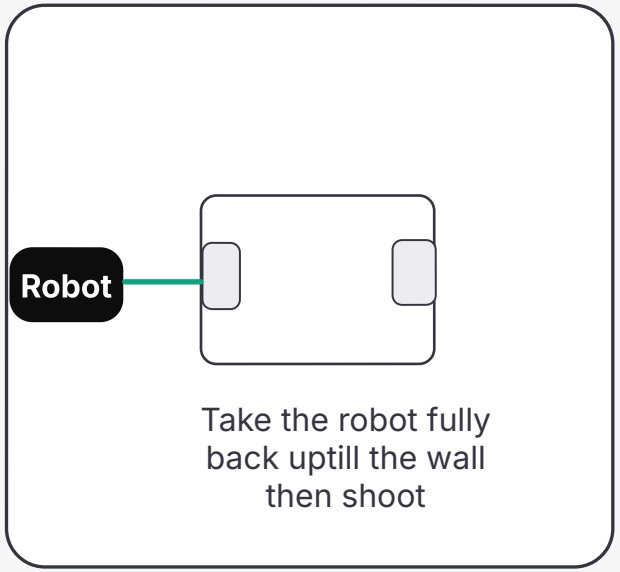
### 1) Use the April-Tags as a reference.




**Remark:** This system is very accurate In-Terms of positioning reliability, but slightly complex for rookie teams.

Unlike the other idea, here we can also configure and control the sideward movement.

### 2) Use the Arena as a reference.




**Remark:** This system is moderately accurate In-Terms of positioning reliability, But easy to build.

But here we can't configure and control the sideward movement.

# Making (V) Constant

**Variables that cause speed changes:**

1) Speed of motor
    a) Battery power reduction

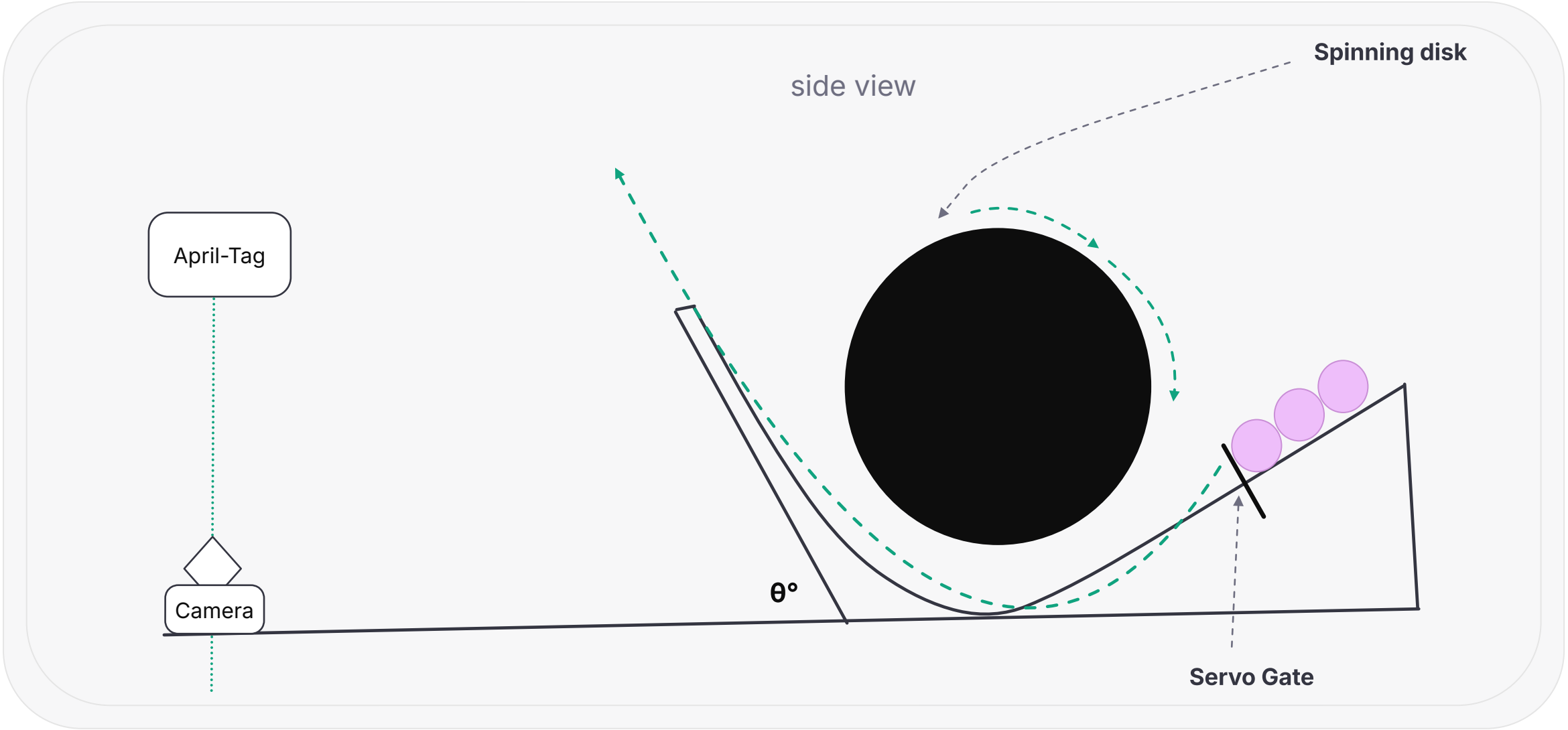
**As per the FTC rules,** we cannot use steppers and also not **EXTERNAL** battery regulation units, but the pre-approved power brick is allowed which also controls voltage.

**Solutions:**

01 **Use a motor.**

02 **Use a battery voltage regulator.**

# Full Shooting mechanism design:



# Logic:

(Apriltag seen)

**Strategy:**

Collect all the balls first, store them in the shooter and then shoot them all together.

**This saves a lot of time.**

01

Align April-Tag to the center by moving right, left, front, back as applicable.

02

**Move back (HD)cms.**

03

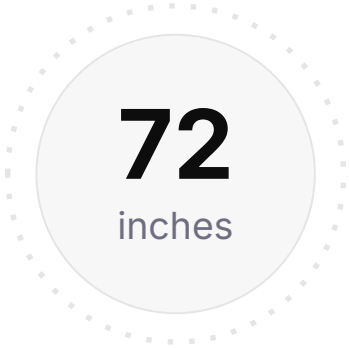
**Shoot!**




## 1. Vertical Displacement & Shooting Distance

**Vertical Displacement (VD):** The height from the floor tiles to the bottom of the Hive cell is 30.6 inches (~77.7 cm / 2.55 ft).

**Horizontal Distance (HD):** Shooting from the alliance boundary to the center-field Hive gives a horizontal distance of 72 inches (6.0 ft / 1.83 m).




## 2. Horizontal Displacement from Hive to Arena Boundary

**Distance:** Since the Hive structure is positioned in the exact center of the field, the horizontal displacement from the start of the Hive to the inner edge of the arena perimeter wall is 72 inches (6.0 ft / 1.83 m).




## 3. Rough Trajectory (RL) & Launch Angle (θ)

**Initial Velocity (v):** 20km/h ~5.56m/s 18.24ft/s.

**Rough Straight-Line Trajectory (Hypotenuse):**

Trajectory (RL) = $\sqrt{(HD^2 + VD^2)}$ = $\sqrt{(72^2 + 30.6^2)}$

RL = ~78.23 inches / ~1.99 m

**Launch Angle θ:** θ = arctan (VD / HD)

**θ = 23.02° rounded-off to "25°"**

# FTC ball Intake mechanism

